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RESEARCH ARTICLE

Rice Variety Identification Based on Transfer Learning Architecture Using DENS-INCEP

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ABSTRACT Rice is a vital staple food for billions of people worldwide, especially in Asia, Africa, and Latin America, where it plays a key role in daily caloric intake and nutrition. With numerous varieties differing in size, shape, colour, texture, and nutritional content, accurate rice variety identification is critical for optimizing production and ensuring food quality. Environmental factors such as soil type and climate further influence these variations, making precise identification essential for improving productivity and reducing waste. However, traditional manual methods of identification, relying on visual characteristics, are prone to human error, resulting in variety mixing, reduced quality, and higher costs. This study addresses these challenges by employing the DENS-INCEP model, a transfer learning approach that integrates DenseNet-201 with the Inception module. DenseNet-201 serves as the backbone for feature extraction, while the Inception module enhances the model's ability to capture multi-scale shape-related features, significantly improving classification accuracy. The model achieved remarkable performance, with an average accuracy of 99.94% across multiple rice varieties. By implementing the DENS-INCEP model, this study contributes to Sustainable Development Goal (SDG) 2 by improving food security through enhanced rice production and supply chain stability. Additionally, it supports SDG 9 by fostering innovation and advancing sustainable agricultural technologies. Furthermore, by reducing errors, waste, and inefficiencies in production and distribution, the model aligns with SDG 12, which emphasizes sustainable consumption and production. Overall, the DENS-INCEP model offers a robust and efficient solution to rice variety identification, addressing global food security challenges while promoting sustainability.

INDEX TERMS DENS-INCEP, DenseNet-201, food security, inception module, process innovation, rice identification, rice variety, transfer learning.

I. INTRODUCTION

Rice holds a foundational role in global agriculture, serving as a primary dietary staple for millions, particularly in Southeast Asia. In Indonesia, where rice constitutes a significant part of the population's daily diet and agricultural economy, accurate classification of rice varieties is essential for optimizing crop management, ensuring quality control, and facilitating efficient distribution. Traditionally, rice variety classification

has relied on manual inspection methods, which are labor-intensive, time-consuming, and susceptible to human error, especially when handling a large number of samples [1], [2], [3]. In addition, visual inspection is often challenged by the high similarity between rice varieties in terms of shape, color, and texture, making it difficult to maintain consistent identification accuracy [4], [5].

The challenge is likely to multiply with the increasing number of rice varieties and their morphological similarity, traditional methods, therefore, struggle to meet the demands precision of rice seed selection. Conventional methodologies

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have frequently encountered challenges in providing reliable outcomes owing to their dependence on manual feature engineering, which is characterized by limited scalability and insufficient adaptability to the diverse spectrum of rice varieties [6], [7]. As a result, there is an urgent need for solutions that are not only efficient and scalable but also capable of high accuracy. This is particularly important in the context of sustainable agriculture, where precise seed identification enabling farmers to select the best varieties that can significantly enhance crop yields and contribute to food security.

The increasing demand for high-precision rice seed classification has prompted research into advanced computer vision techniques. These efforts focus on developing automated systems capable of accurately identifying and categorizing rice seed varieties based on morphological traits such as size, texture, shape, and color. For instance, prior studies [8], [9], [10] implemented image processing methods to classify rice varieties based on geometric and morphological features extracted from 2D seed images. In the same vein, researchers in computer vision have also explored the application of RGB imaging, often in conjunction with other methods, to automate and improve the seed identification process [11], [12]. While these early image classification approaches have shown promise in rice seed identification, they have often struggled to deliver consistent results due to the inherent variability in rice seed morphology, environmental conditions, and imaging setups [4], [5], [13]. Similarly, challenges remain in RGB, particularly in ensuring consistent classification across different laboratories and conditions [14].

In subsequent investigations, the emphasis transitioned towards augmenting the accuracy of classification via the incorporation of sophisticated feature extraction techniques. Scholars commenced the examination of texture analysis as a pivotal attribute for the identification of rice seed varieties. For instance, researchers [15], [16] employed Grayscale Co-Occurrence Matrices (GLCM) to elucidate texture-based attributes that distinguished between rice seed types. These inquiries illustrated that texture analysis could serve as a valuable complement to shape- and size-oriented features, thereby enhancing classification efficacy. Nonetheless, the main limitations of GLCM for classification include its time-consuming nature and lower accuracy in regions near class borders that is less appropriate for practical implementations in real-world scenarios [17].

The advent of Deep Learning (DL) and Machine Learning (ML) instigated a transformative shift in the domain of seed classification. Advanced computer vision techniques, particularly those leveraging DL, have addressed these limitations by enabling models to learn hierarchical and intricate features directly from raw images. Unlike previous approaches that relied on manually crafted features, DL architectures, possess the capability to autonomously acquire hierarchical features directly from unprocessed images. Research has demonstrated that the amalgamation of image processing

methodologies with various DL frameworks can significantly improve the precision of rice seed classification, attaining cutting-edge outcomes that exceed those of conventional methods, thus promoting enhanced agricultural management and more accurate yield forecasting in farming practices [18], [19], [20].

Many studies have explored the application DL and ML for classifying objects, patterns, and features. Similarly, rice variety identification can be automated through detailed feature extraction of rice image, which can be performed automatically using efficient image recognition techniques. For instance, Koklu et al. [21] achieved 99.87% accuracy in classifying five rice varieties (Arborio, Basmati, Ipsala, Jasmine, Karacadag) using 75,000 images and Artificial Neural Networks, demonstrating the high performance of deep learning in rice classification. Likewise, Saxena et al. [22] classified five global rice varieties using various algorithms, with Random Forest achieving the highest accuracy of 99.85%, followed by Decision Tree at 99.68%.

While traditional ML and DL approaches have demonstrated high accuracy, they often require manual feature extraction and extensive training. To overcome these limitations, Transfer Learning (TL) has emerged as an effective method for classification and identification tasks by leveraging pre-trained models. Chen et al. [23] in their study on rice leaf disease detection showed that the proposed DENS-INCEP model an average prediction accuracy of no less than 94.07% on a public dataset. Furthermore, when multiple diseases were considered, the model achieved an impressive average accuracy of 98.63% for classifying rice disease images. These findings indicate that DENS-INCEP is highly effective for complex image classification tasks, such as plant disease identification, and can also be adapted for other applications, including rice variety identification and other related challenges. The DENS-INCEP model combines DenseNet-201 and the Inception Module. DenseNet-201 acts as the backbone and feature extractor, while the Inception Module is added to improve performance and computational efficiency [23].

Additionally, a number of recent studies have also demonstrated high accuracy in disease identification using deep learning models. For example, Saleem et al. [24] proposed a hybrid deep learning architecture that enhances multi-crop leaf disease detection by integrating vegetative feature isolation techniques, achieving superior classification performance. Similarly, Naeem et al. [25] introduced a deep learning-based approach for potato leaf disease identification, utilizing wrapper feature selection and feature concatenation to improve classification accuracy. Additionally, Saleem et al. [26] developed a multi-scale feature extraction and fusion method for wheat disease classification, further highlighting the effectiveness of deep learning in plant disease identification.

While disease identification is crucial for mitigating crop loss, seed identification plays an equally essential role in optimizing crop yield leading to improving agriculture

TABLE 1. Accuracy comparison of different deep learning models.

Study	Model	Accuracy (%)
Iqbal et. al. [28]	NasNetMobile	99.20
Farahnakian et. al. [6]	EfficientNetB0	99.67
Hazra et. al. [29]	InceptionV2	99.80
Prasetyo [30]	Xception	99.81
Kumar et. al. [31]	DarkNet	99.75
Cheng [32]	VGG16	92.00
Prasetyo [33]	ResNet50	98.81
	ResNet101	98.75
Vania et. al. [2]	MobileNetV2	96.40

productivity. A considerable amount of research has studied rice identification and classification based on different features, with most studies focusing on color-based features rather than shape [27]. Although this approach is effective for distinguishing rice varieties with distinct color differences, it may not be sufficient for varieties with overlapping color characteristics.

Recent advancements have shifted towards shape-based classification, which can be more robust under varying lighting conditions. Further review identified eight studies that specifically explored rice variety classification based on shape-related features using deep learning models making it a promising approach for precision agriculture. Table 1 presents a comparative analysis of eight state-of-the-art deep learning models applied to rice variety classification using the Kaggle Rice Image Dataset (75,000 samples).

This study, therefore, aims to demonstrate the effectiveness of the DENS-INCEP model in rice variety classification by leveraging its capability to capture multi-scale shape-related features. To validate the performance, comparisons are made with various deep learning architectures, including NasNetMobile, EfficientNet-B0, InceptionV2, Xception, DarkNet, ResNet50, ResNet101, MobileNetV2, and VGG16, which have been discussed and summarized in Table 1. These comparisons provide a benchmark, highlighting the strengths the DENS-INCEP model against these state-of-the-art architectures. By enabling faster and more accurate rice variety identification, this approach helps optimize crop management, streamline distribution, and reduce misclassification risks, ultimately supporting sustainable agricultural practices and food security initiatives [34].

The motivation behind this research aligns with several of the United Nations' Sustainable Development Goals (SDGs). By enhancing the accuracy and efficiency of rice classification, our model contributes to SDG 2: Zero Hunger by promoting sustainable agricultural practices that optimize crop yield and quality. The model also supports SDG 12: Responsible Consumption and Production by enabling better traceability and quality assurance in the rice supply chain, which is essential for maintaining standards in both domestic and export markets. Additionally, by creating technology that can be applied in rural and resource-limited settings, this research aligns with SDG 9: Industry, Innovation, and Infrastructure, advancing digital innovation

in the agricultural sector and supporting technological accessibility.

II. LITERATURE REVIEW

This section reviews the relevant literature on rice variety classification, highlighting the advancements in computer vision and deep learning techniques applied to agricultural domains. Rice variety identification is critical for optimizing production, ensuring quality control, and supporting sustainable agriculture. Traditional classification methods face limitations due to variability in grain morphology and environmental factors. Recent advancements in transfer learning have enabled the use of pre-trained models, such as DenseNet-201 and the Inception module, for extracting hierarchical and multi-scale features, improving classification accuracy. The integration of these models, referred to as the DENS-INCEP architecture in this study, represents a significant step forward in addressing the challenges of fine-grained classification.

A. RICE VARIETY

Rice is the grain of the rice plant (*Oryza sativa*), is one of the most important staple crops globally [35], [36]. It serves as a primary food source for billions of people, particularly in regions such as Asia, Africa, and Latin America [37]. In developing countries, rice plays a vital role in fulfilling daily caloric needs. For instance, it provides approximately 50% of daily calorie intake and is a significant source of protein for around 520 million individuals living in poverty across Asia [38]. Composed of approximately 80% carbohydrates, 12% water, and 7% protein, rice is an efficient source of energy. However, its nutritional content can vary depending on the specific variety of rice [37].

Rice is broadly classified into two main subspecies: Japonica and Indica [39]. Japonica varieties are generally shorter, rounder, and stickier when cooked, often cultivated in temperate regions. In contrast, Indica varieties are typically long, slender, and less sticky, thriving in tropical climates. These differences in grain structure and growing conditions contribute to the diversity in nutritional content and cooking properties found across various rice varieties [40], [41].

Each rice variety exhibits distinct visual characteristics, and these traits are influenced by the subspecies to which they belong. For instance, Indica varieties, such as Basmati, are part of the Indica subspecies and are known for their long, slender grains, with subtle differences in texture and aroma. Similarly, Japonica varieties, which belong to the Japonica subspecies, are characterized by their short, round grains. These differences arise from a combination of genetic and environmental factors, reflecting the plant's adaptation to local conditions and agricultural practices. Furthermore, cultivation methods influence both the nutritional quality and productivity of rice, highlighting the importance of selecting the appropriate variety to meet the specific needs of populations in different regions [42].

B. TRANSFER LEARNING

TL is a DL technique that utilizes knowledge from one or more source domains to improve the performance of a target learner. This technique typically involves using a pre-trained neural network model, such as those trained on large datasets like ImageNet, as a foundation for tackling new tasks [43]. ImageNet is a large-scale dataset designed to support advancements in visual recognition and machine learning. It comprises over 14 million labeled images organized within the WordNet hierarchy, offering a broad and deep range of categories that make it an essential resource for training and evaluating computer vision models [44].

The core principle of TL is leveraging a trained model as an initial foundation. Such models have learned to recognize basic features like edges, shapes, and textures through the ImageNet dataset. Utilizing this prior knowledge saves time and computing resources, as the model doesn't require training from scratch, and it improves accuracy given its strong foundational understanding [45].

Figure 1 illustrates the TL process, where a pre-trained model is adapted for a new task. Initially, the model is trained on ImageNet, extracting essential features. These features are then leveraged for predictions through the model's head. When the model is applied to a new task, it uses the same architecture but with a new head, which is retrained to adapt to the target dataset. This process of transferring pre-existing knowledge enables faster learning, especially when the new dataset is smaller or has fewer labelled data points, making TL a highly effective technique in DL.

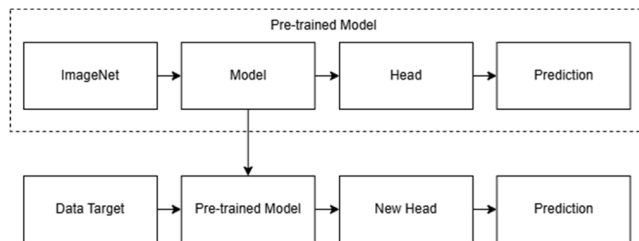


FIGURE 1. Transfer learning process.

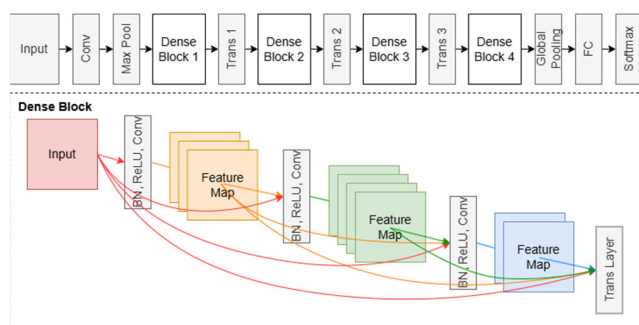


FIGURE 2. DenseNet-201 architecture.

C. DenseNet-201

DenseNet-201 is a variant of DenseNet (Densely Connected Convolutional Networks) introduced by Huang et al. [46].

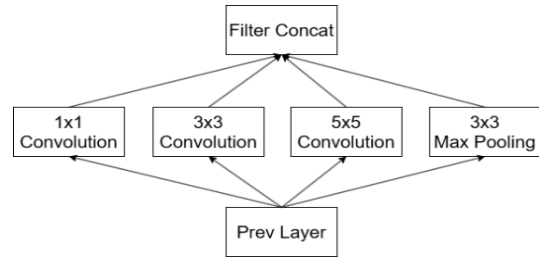


FIGURE 3. Inception module.

This model, with its 201 convolutional layers, was designed to address the vanishing gradient problem often encountered in very DL models, where gradients diminish as they move backward through the layers, hindering effective learning in earlier layers [47]. The core feature of DenseNet lies in its dense connections, where each layer is directly connected to every other layer within the same dense block.

Figure 2 illustrates the DenseNet-201 architecture, that consists of four dense blocks connected by transition layers. Each block is connected to every other layer within the same block, facilitating efficient feature reuse and reducing the vanishing gradient problem. Between these blocks, transition layers are used to down-sample the feature maps, allowing the model to manage computational complexity while maintaining a high level of accuracy.

This architecture enables a highly efficient flow of information and gradients, as each layer has access to the feature maps of all preceding layers within the block. This dense connectivity not only helps to alleviate vanishing gradients but also reduces redundancy, as features learned by earlier layers are reused rather than relearned in later layers. Consequently, the DenseNet-201 model requires fewer parameters and improves both computational efficiency and image classification accuracy.

D. INCEPTION MODULE

The Inception Module was first introduced in the GoogLeNet architecture by Szegedy et al. [48]. This module was specifically designed to enhance computational efficiency and prevent overfitting in deep networks, addressing some common challenges of DL architectures. The Inception Module utilizes multiple convolution operations with varying kernel sizes 1×1 , 3×3 , and 5×5 executed in parallel within a single layer.

Figure 3 depicts the Inception Module architecture, which processes multiple convolutional operations with different kernel sizes (1×1 , 3×3 , and 5×5) in parallel within a single layer. This parallel approach enables the model to capture features at various scales, detecting both fine details and broader patterns in the input image. By combining multiple receptive fields within the same layer, the Inception Module significantly improves the model's ability to learn complex spatial patterns without requiring excessive depth, which helps maintain computational efficiency.

TABLE 2. DENS-INCEP main parameter.

Module	Function	Output Shape	Kernel Size	Stride
Input	None	(None, 224, 224, 3)	None	None
Dense Block1	Convolution	(None, 56, 56, 256)	$1 \times 1, 3 \times 3$	1
Trans Layer1	Convolution, Pooling	(None, 28, 28, 128)	$1 \times 1, 2 \times 2$	1, 2
Dense Block2	Convolution	(None, 28, 28, 512)	$1 \times 1, 3 \times 3$	1
Trans Layer2	Convolution, Pooling	(None, 14, 14, 256)	$1 \times 1, 2 \times 2$	1, 2
Dense Block3	Convolution	(None, 14, 14, 1792)	$1 \times 1, 3 \times 3$	1
Trans Layer3	Convolution, Pooling	(None, 7, 7, 896)	$1 \times 1, 2 \times 2$	1, 2
Inception Module	Inception Module	(None, 512)	$1 \times 1, 3 \times 3, 5 \times 5$	None
FC Layer	Fully-connect	(None, 512)	None	None
Output	Softmax Regression	(None)	Classifier	None

E. DENS-INCEP

DENS-INCEP, first introduced by Chen et al. [23] in 2020, is a neural network model that integrates two well-known architectures: the pre-trained DenseNet-201 and the Inception Module. DenseNet-201 serves as the primary feature extractor. Meanwhile, the Inception Module focuses on multi-scale feature extraction. For example, the 1×1 convolution filters capture fine-grained details like the sharpness of the grain edges, while the 3×3 and 5×5 filters capture broader shape characteristics such as elongation or roundness. This fusion of DenseNet-201 with the Inception Module creates a novel approach to TL architecture, enabling improved model performance, particularly in tasks where feature extraction from multiple scales is crucial.

DENS-INCEP is built on the pre-trained DenseNet-201 model, with frozen layers to preserve the rich features learned from a large-scale dataset. The architecture utilizes three dense blocks from DenseNet-201 as the core feature extractor, followed by an Inception Module to capture multi-scale features and improve pattern recognition. Then a Global Average Pooling (GAP) layer reduces spatial dimensions while retaining key information, this is followed by a fully connected (FC) layer with 502 neurons and Rectified Linear Unit (ReLU) activation for enhanced learning, and a final FC layer with Softmax activation to output class probabilities for accurate classification.

Figure 4 showing the DENS-INCEP architecture, which includes three dense blocks and an Inception module. The architecture begins by processing input image data through a convolutional layer to extract initial features. Each dense block connects every convolutional layer to all previous layers, and after each dense block, a transition layer reduces data dimensions through convolution and pooling.

After passing through the three dense blocks, the architecture moves to the Inception module, which contains parallel paths with different filter sizes ($1 \times 1, 3 \times 3, 5 \times 5$) to capture

features at multiple scales. The outputs from these paths are combined through concatenation, resulting in a richer feature representation. Finally, a Global Average Pooling (GAP) layer reduces dimensions, followed by a FC layer and a Softmax function, which provides probability predictions for each class. This combination is designed to enhance both accuracy and efficiency in image classification tasks. Details of the main parameters of DENS-INCEP are summarised in Table 2.

Table 2 shows the main parameters of the DENS-INCEP network, including the various modules and functions applied at each stage of the architecture. It lists the output shape, kernel size, and stride used for each module, from the initial feature extraction in the Dense Blocks to the final classification using Softmax Regression. This provides an overview of how data is processed through the network.

F. EVALUATION METRICS

Evaluation metrics are quantitative measures used to assess the performance of a classification model. They provide a comprehensive understanding of how well the model predicts classes, helping to identify its strengths and weaknesses.

1. **Confusion matrix** is a tool for evaluating the performance of a classification model [49]. It provides a detailed breakdown of the model's predictions by categorizing them into four key components: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). An example confusion matrix can be seen in Figure 5 illustrating the distribution of TP, TN, FP, and FN in the classification results. This matrix provides a detailed overview of the model's prediction performance for each class, highlighting both correct classifications and the types of errors made.
2. **Receiver operating characteristic (ROC) curve** is an analytical method used to evaluate the performance of a classification model [50]. It does so by illustrating the trade-off between the true positive rate (TPR) and the false positive rate (FPR) at various threshold levels. By plotting these rates, the ROC curve helps in understanding the model's classification ability and its sensitivity to various decision thresholds. It is a powerful tool for comparing models and selecting the optimal threshold for maximizing classification performance. TPR can be calculated using (1), and FPR is determined by (2).

$$TPR = \frac{TP}{TP + FN} \quad (1)$$

$$FPR = \frac{FP}{FP + TN} \quad (2)$$

3. **Accuracy** is a metric used to measure the overall performance of a classification model [51]. Accuracy indicates the proportion of correct predictions out of all predictions made by the model. Accuracy can be

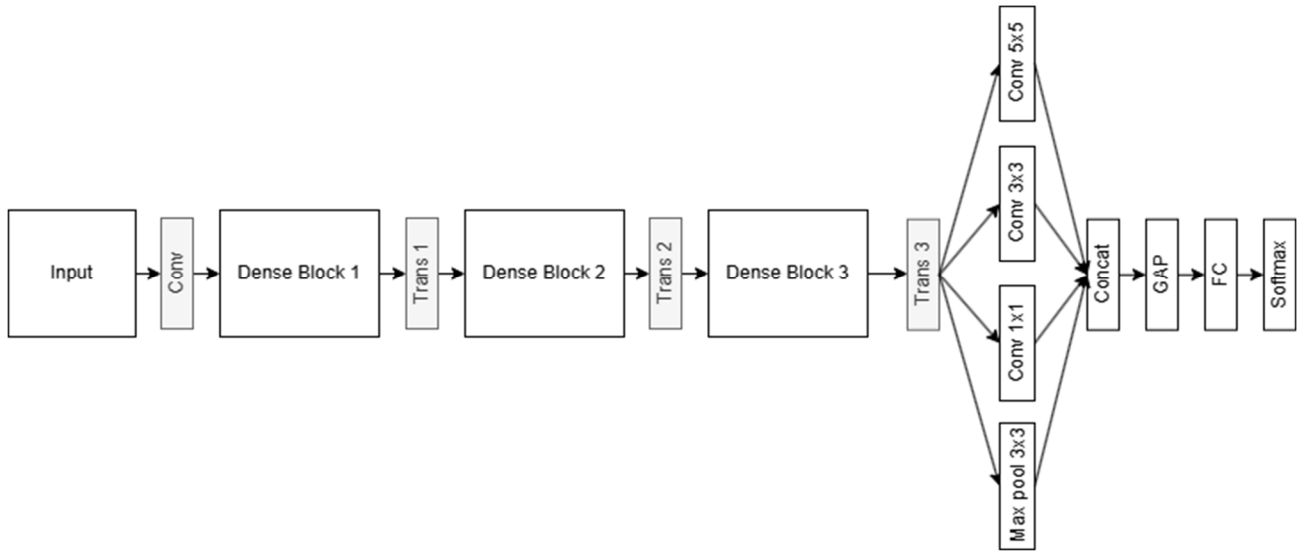


FIGURE 4. DENS-INCEP architecture.

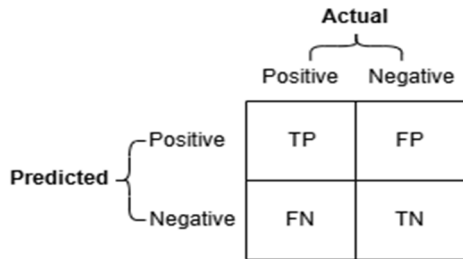


FIGURE 5. Confusion matrix illustration.

calculated using (3).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

4. **Sensitivity**, also known as Recall or TPR, is a metric used to evaluate a classification model's ability to correctly identify positive instances. Sensitivity indicates how well the model detects actual positive cases without any missing. Sensitivity can be calculated using (4).

$$Sensitivity = \frac{TP}{TP + FP} \quad (4)$$

5. **Specificity** is a metric used for evaluating a classification model's ability to correctly identify negative instances. Specificity measures how effectively the model avoids false alarms by accurately predicting negative cases. Specificity This can be calculated using (5).

$$Specificity = \frac{TN}{TN + FP} \quad (5)$$

III. MATERIAL AND METHOD

This section outlines the research stages undertaken to achieve accurate rice variety identification. The overall

research workflow is illustrated in Figure 6, which provides a visual representation of the study's process. The workflow begins with the input dataset and progresses through several key stages, including data preprocessing, dataset splitting, model training, and evaluation. Each of these steps is critical to ensuring the accuracy and reliability of the results. The subsequent subsections will provide a detailed explanation of each stage in the workflow.

A. DATASET AND TOOLS

The data used in this study primarily consists of 5 varieties of rice grain images. The dataset contains 75,000 images of rice grain obtained from shared dataset that is publicly available at Kaggle Rice Image Dataset. The rice varieties are categorized into five classes: Arborio, Basmati, Ipsala, Jasmine, and Karacadag, each image has a resolution of 250×250 pixels. Figure 7 shows samples of different rice varieties used in this study, including (a) Arborio, (b) Basmati, (c) Ipsala, (d) Jasmine, and (e) Karacadag. Each rice variety exhibits distinct visual characteristics, such as differences in grain shape, size, and color.

For instance, Arborio grains are typically short and round, while Basmati grains are known for their long and slender appearance. Ipsala grains are medium-sized, whereas Jasmine rice has long, slightly curved grains. Karacadag rice features a distinctive grain shape that sets it apart from the others. Each class consists of 15,000 images, these were then split into two subsets at a ratio of 70:30.

Table 3 shows the distribution of the rice variety dataset used in this study. Each category consists of 15,000 images, with the training set contains 10,500 images, while the testing set includes 4,500 images, ensuring a balanced dataset for both model training and evaluation. The tools used in this study include a computer, the Python programming language,

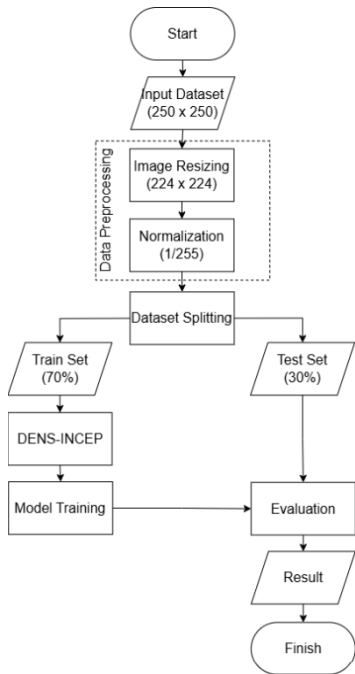


FIGURE 6. Research workflow.

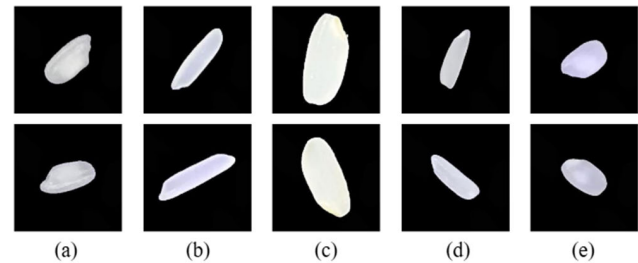


FIGURE 7. Rice variety samples (a) Arborio, (b) Basmati, (c) Ipsala, (d) Jasmine, (e) Karacadag.

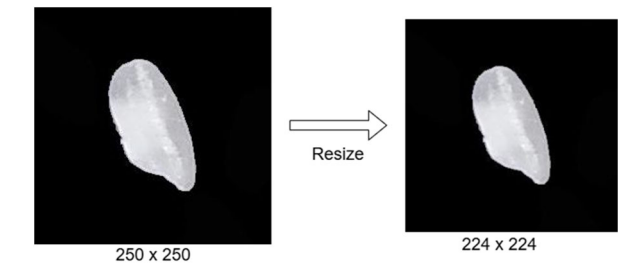


FIGURE 8. Image resizing.

and Google Colab, with the program running on a T4 GPU. Model development uses the TensorFlow framework and Keras library, which provides essential functions for building, training, and evaluating models.

B. DATA PREPROCESSING

Data preprocessing ensures the quality and consistency of the input data before it is fed into the model. In this study, preprocessing steps include resizing images to a uniform dimension

TABLE 3. Data distribution.

Category	Number of Image	Traning Set	Testing Set
Arborio	15,000	10,500	4,500
Basmati	15,000	10,500	4,500
Ipsala	15,000	10,500	4,500
Jasmine	15,000	10,500	4,500
Karacadag	15,000	10,500	4,500

and normalizing pixel values to improve convergence during training.

1. **Image Resizing** is a crucial preprocessing step that ensures all images conform to the required input dimensions of the DENSE-INCEP model. In this study, the original rice grain images, which have a resolution of 250×250 pixels, are resized to 224×224 pixels to align with the input requirements of DenseNet-201, which serves as the backbone of our model. This adjustment ensures that the pretrained model's feature extraction capabilities are effectively leveraged without requiring modifications to the network's structure. A similar approach was employed in Saleem et al. [26], where images were transformed into greyscale and resized to 224×224 pixels to standardize input dimensions and enhance model efficiency. By maintaining a uniform image size, the resizing process optimizes computational efficiency, facilitates robust feature extraction, and preserves key morphological characteristics essential for distinguishing rice varieties. Figure 8 illustrates the image resizing process, showing how the original 250×250 pixel images are downscaled to 224×224 pixels while preserving key morphological features necessary for classification. This resizing serves multiple purposes. First, it ensures consistency across the dataset by standardizing input dimensions, preventing discrepancies that could arise from varying image sizes, and ensuring uniformity in feature extraction. Second, it enhances computational efficiency by reducing image size, which optimizes memory usage and processing speed, making the model more efficient during both training and inference. Lastly, the resizing process is designed to preserve visual information, particularly essential shape-related features such as grain contour, edge sharpness, and curvature, which are crucial for distinguishing between different rice varieties. By enforcing a standardized input size, this preprocessing step enhances the DENSE-INCEP model's ability to learn and recognize the distinguishing characteristics of different rice varieties, ultimately contributing to improved classification accuracy.
2. **Data Normalization** is a crucial preprocessing step that ensures consistent pixel intensity values across all input images, improving model convergence and stability during training. In this study, the original pixel

values of rice grain images range from 0 to 255, which can create imbalances during weight updates in deep learning models. Features with larger numerical values may disproportionately influence the learning process, leading to inefficient training and slower convergence. To mitigate this issue, min-max normalization is applied by scaling pixel values to a standardized range of 0 to 1 using the transformation:

$$x_{\text{normalized}} = \frac{x}{255} \quad (6)$$

where x represents the original pixel intensity. By normalizing pixel values, the model benefits in several ways. First, it enhances numerical stability, preventing large gradient updates that could cause erratic weight adjustments. Second, it accelerates convergence, as deep learning models typically perform better when input features are within a small, uniform range. Third, it reduces sensitivity to variations in lighting conditions, ensuring that the model learns robust shape-based features rather than being influenced by absolute pixel intensity differences. An example of this normalization process is illustrated in Figure 9, which shows a comparison of pixel intensity distributions before and after normalization. By applying this technique, the model can effectively learn and generalize key morphological characteristics of rice varieties without being biased by variations in pixel intensity values.

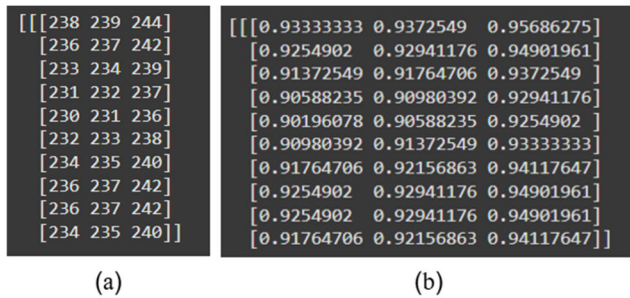


FIGURE 9. Data normalization (a) Pixel array before normalization, (b) pixel array after normalization 9.

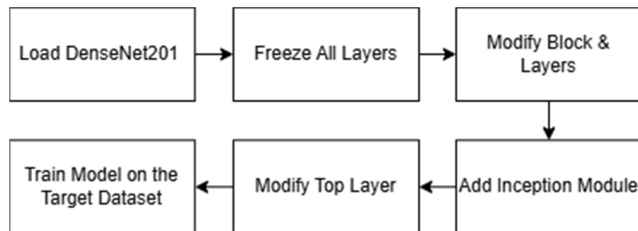


FIGURE 10. DENS-INCEP architecture design.

C. DENS-INCEP ARCHITECTURE DESIGN

The design of the DENS-INCEP model follows a structured process to optimize its performance for rice variety classification. Figure 10 illustrates the architecture design, detailing

the steps involved in adapting the DenseNet-201 backbone with the Inception module to enhance feature extraction. The model construction follows these key steps:

1. Initialize the network parameters using the pre-trained DenseNet-201 model, leveraging its feature extraction capabilities from the ImageNet dataset.
2. Adapt the model to the new classification problem by freezing the weights of the pre-trained layers to retain essential learned features.
3. Remove the final fully connected (FC) layer originally designed for ImageNet classification to allow for custom feature learning specific to rice variety identification.
4. Integrate the Inception module to enhance multi-scale feature extraction, improving the model's ability to capture intricate shape variations in rice grains.
5. Add a new FC layer with ReLU activation, consisting of 512 neurons, followed by a Softmax output layer to classify rice varieties into distinct categories.

To ensure optimal training performance, we have fine-tuned the model using the following hyperparameters:

- Optimizer: Adaptive Moment Estimation (Adam)
- Learning Rate: 0.001
- Batch Size: 64
- Epochs: 30
- Loss Function: Categorical Crossentropy
- Activation Functions: ReLU (hidden layers), Softmax (output layer)

These hyperparameter selections balance training stability, convergence speed, and generalization, ensuring the DENS-INCEP model achieves high accuracy while maintaining computational efficiency.

D. MODEL TRAINING AND TESTING

At this stage, the DENS-INCEP model is trained using the pre-processed rice grain dataset and benchmarked against other pretrained models, including VGG19, ResNet-50, DenseNet-201, and InceptionV3. The training process follows a structured approach to optimize model performance while ensuring robustness and efficiency. The training process consists of the following key components:

1. **Forward Propagation:** The input images are passed through each layer of the DENS-INCEP architecture, extracting multi-scale shape features and generating prediction outputs.
2. **Backward Propagation:** The error between the predicted and actual labels is computed using the Categorical Crossentropy loss function. The network weights are updated using the Adam optimizer with a learning rate of 0.001 to improve convergence.
3. **Evaluation and Overfitting Prevention:** The model's performance is monitored at each epoch using validation data. Training is conducted with a batch size of 64 over 30 epochs, ensuring a balance between learning efficiency and computational feasibility.

After training, performance metrics including accuracy, loss, sensitivity, specificity, Confusion Matrix, and ROC Curve are calculated to assess the model's effectiveness in distinguishing rice varieties. To efficiently train and evaluate the DENS-INCEP model, the following hardware and software environment was used:

- Hardware: NVIDIA Tesla T4 GPU, 16GB RAM
- Software: TensorFlow, Keras, Python (Google Colab environment)
- Training Time: Approximately 3 hours per model
- Inference Time: <1 second per image, making the model suitable for real-time classification applications

These computational resources were crucial in achieving high-performance training and inference. The GPU acceleration significantly reduced model training time, allowing for deeper network architectures such as DenseNet-201 and Inception to be efficiently fine-tuned without excessive delays. The fast inference time ensures that the trained model can be deployed in practical agricultural settings, enabling rapid and accurate rice variety classification.

In addition, to ensure the practical deployment feasibility of the DENS-INCEP model, we conducted a comprehensive performance profiling analysis. The model was tested on an NVIDIA Tesla T4 GPU with 15.36 GB of available memory to assess its computational efficiency, including GPU memory usage, inference time, and model size. The performance profiling was conducted on an NVIDIA Tesla T4 GPU, a commonly used hardware for cloud-based and edge computing applications. The evaluation focused on GPU memory usage, inference time, and model size to assess the model's efficiency and deployability. GPU memory usage was monitored using the NVIDIA System Management Interface (nvidia-smi) tool during training and inference to measure the maximum memory utilized. Inference time was evaluated by processing 352 batches of input images, calculating the total time taken and the average inference time per batch to determine real-time applicability.

Finally, to comprehensively evaluate the performance of the DENS-INCEP model, we conducted a benchmarking study against multiple state-of-the-art deep learning architectures. The selected models include NasNet-Mobile, EfficientNet-B0, InceptionV2, Xception, DarkNet, ResNet50, ResNet101, MobileNetV2, and VGG16 as summarized in Table 1. These architectures were chosen based on their established effectiveness in prior research utilizing the same dataset and their varying trade-offs between accuracy, computational complexity, and inference efficiency.

IV. RESULT AND DISCUSSION

This section presents the findings of the study and provides an in-depth analysis of the results obtained from the proposed DENS-INCEP model for rice variety classification. The performance of the model is evaluated based on key metrics, such as accuracy, precision, recall, and F1-score, to demonstrate its effectiveness. Comparisons with existing methods

TABLE 4. Comparison result.

Model	Training Accuracy	Testing Accuracy	Training Loss	Testing Loss
VGG19	0.9912	0.9910	0.0241	0.0278
ResNet50	0.9801	0.9732	0.0560	0.0739
DenseNet201	0.9995	0.9966	0.0015	0.0127
InceptionV3	0.9985	0.9939	0.0043	0.0281
DENS-INCEP	0.9996	0.9986	0.0012	0.0049

are also discussed to highlight the improvements achieved. Additionally, the implications of the results are analyzed in the context of practical applications, including their relevance to food security and sustainable agricultural practices. The findings are further examined to understand their alignment with the study objectives and their contribution to addressing the challenges of rice variety identification.

A. RESULT

The experiment involves conducting trials using the DENS-INCEP model and comparing its performance with conventional TL models such as VGG19, ResNet50, DenseNet-201, and InceptionV3. All models are trained with the same parameters, to ensure consistency in the evaluation process, allows for a fair comparison of model performance across different architectures in identifying rice varieties.

Based on Table 4, the DENS-INCEP model demonstrates superior performance in classifying rice variety images. The model achieves the highest metrics among all models, with a Training Accuracy of 99.96%, Testing Accuracy of 99.86%, Training Loss of 0.12%, and Testing Loss of 0.49%. These results indicate that the DENS-INCEP model outperforms other architectures, such as VGG19, ResNet-50, DenseNet-201, and InceptionV3, in both accuracy and loss metrics. Furthermore, the confusion matrix and ROC curve of predicting results can be seen in Figure 11 and Figure 12, and the corresponding performance metrics of each class are calculated in Table 5.

As shown in Figure 11, the confusion matrix highlights the model's strong ability to accurately distinguish rice varieties, with most predictions aligning correctly along the diagonal. The model successfully classified 4498 Arborio, 4492 Basmati, 4500 Ipsala, 4486 Jasmine, and 4492 Karacadag samples. However, some misclassifications occurred, including 2 Arborio samples misclassified (1 as Jasmine and 1 as Karacadag), 8 Basmati samples misclassified as Jasmine, 14 Jasmine samples misclassified as Arborio, and 8 Karacadag samples misclassified as Arborio.

These misclassifications could be attributed to the visual similarities in shape features between certain rice varieties, particularly Jasmine and Arborio, which may share overlapping morphological characteristics. Additionally, Basmati and Jasmine misclassifications suggest that the model may have encountered challenges in distinguishing fine-grained structural differences between these varieties. The limited number of misclassified samples indicates that while the model performs exceptionally well, further refinements—

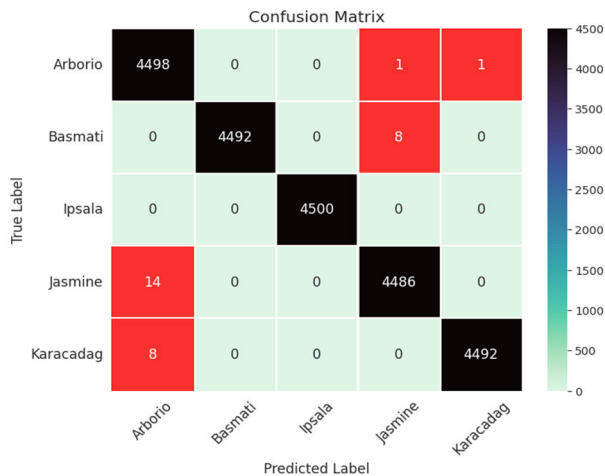


FIGURE 11. Confusion matrix.

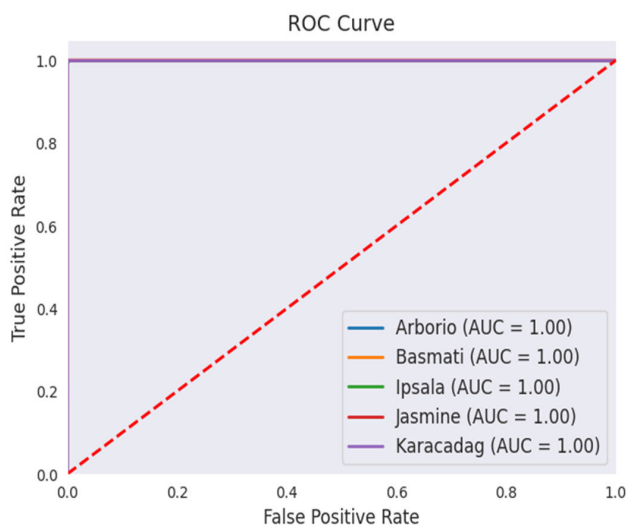


FIGURE 12. ROC curve.

such as enhancing feature extraction, incorporating additional discriminative layers, or expanding training data with more diverse environmental variations—could further reduce classification errors.

This observation is further supported by the ROC curve shown in Figure 12, where all rice varieties exhibit near-perfect classification performance with an AUC of 1.00 for each class. Since all rice varieties achieved an AUC of 1.00, their ROC curves overlap perfectly, resulting in a single line representation. Due to this, visual differentiation among the curves is not possible, but the legend distinctly identifies each rice variety. This result highlights the model's exceptional ability to accurately distinguish between all classes, reinforcing its effectiveness in rice variety classification. To improve accuracy, future work could explore additional pre-processing techniques, adaptive weighting for misclassified classes, or fine-tuning the Inception module to

better capture intricate shape differences. Despite these minor misclassifications, the overall performance demonstrates the robustness and reliability of the DENS-INCEP model in rice variety classification.

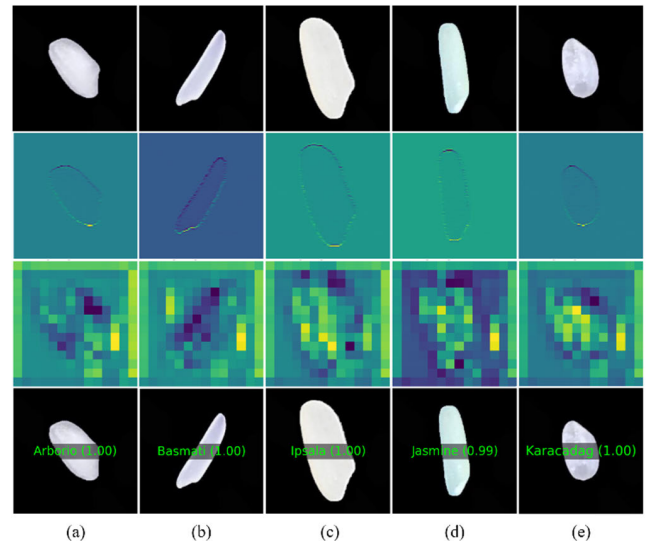


FIGURE 13. Prediction samples (a) Arborio, (b) Basmati, (c) Ipsala, (d) Jasmine, (e) Karacadag.

Table 5 provides a comprehensive evaluation of the model's performance in classifying different rice varieties, demonstrating consistently high accuracy across all classes, with an overall average accuracy of 99.94%. Ipsala exhibited the most outstanding performance, achieving 100% accuracy, sensitivity, and specificity, indicating flawless classification without any misclassification instances. Basmati and Karacadag also demonstrated strong classification results, both attaining 99.96% accuracy, with Basmati's sensitivity at 99.82% and Karacadag's specificity at 99.99%, reinforcing the model's robustness in precise identification.

While Arborio recorded 99.89% accuracy, it had the lowest specificity at 99.88%, suggesting a slightly higher false positive rate compared to other varieties. Similarly, Jasmine achieved 99.90% accuracy, with 99.69% sensitivity, indicating strong but slightly less optimal recall for this variety. Overall, the model maintained 99.86% sensitivity and 99.96% specificity, underscoring its exceptional reliability in distinguishing rice varieties with minimal misclassification errors. These findings confirm the DENS-INCEP model's superior feature extraction and classification capabilities, making it highly suitable for rice variety identification applications.

As illustrated in Figure 13, the top row presents the original input images of different rice varieties, namely Arborio, Basmati, Ipsala, Jasmine, and Karacadag. The second row displays the corresponding feature maps extracted from the initial convolutional layer (conv1_conv) of the DenseNet-201 architecture. This layer captures low-level structural features such as edges, contours, and textures, forming

TABLE 5. Performance metrics.

Variety	Pred	Correct	Accuracy	Sensitivity	Specificity
Arborio	4500	4498	99.89	99.96	99.88
Basmati	4500	4492	99.96	99.82	100
Ipsala	4500	4500	100	100	100
Jasmine	4500	4486	99.90	99.69	99.95
Karacadag	4500	4492	99.96	99.82	99.99
Average	-	-	99.94	99.86	99.96

the foundational elements for subsequent feature extraction. The third row shows the feature maps obtained from the conv4_block48_concat layer, which integrates information from multiple convolutional paths within the Inception module. This layer captures more abstract and high-level representations of the rice grains by analyzing their multi-scale shape-related features. Finally, the bottom row presents the model’s classification output, including the predicted rice variety and its associated probability value. The consistent and accurate predictions across all samples demonstrate the model’s robustness in distinguishing rice varieties based on shape-related features.

Based on the analysis of Figure 13, the predicted classes for all rice samples align perfectly with their actual rice variety labels, confirming the model's high accuracy in distinguishing different rice varieties. The feature visualization across multiple layers illustrates the progressive feature extraction process of the DENS-INCEP architecture. The initial convolutional layers capture fundamental textures and edges, while deeper layers within the Inception module extract more abstract, high-level shape-related features. This demonstrates the model's robustness in feature representation and classification, ensuring precise identification of rice varieties based on their unique visual characteristics.

B. DISCUSSION

The proposed approach architecture of DENS-INCEP, as shown previously in Figure 4, combines DenseNet-201 with the Inception module to identify rice varieties. This combination was chosen because DenseNet-201, with its dense connections, provides rich and diverse feature representations, while the Inception module enhances the model’s ability to capture features at multiple scales. To evaluate its performance, DENS-INCEP was compared with several well-known models, including VGG19, ResNet-50, DenseNet-201, and InceptionV3. All models were trained and tested on the same dataset with identical settings to ensure a fair comparison.

As summarized in Table 4, DENS-INCEP achieved the highest accuracy and the lowest loss in both training and testing, it’s outperforming the other models. For instance, DenseNet-201 achieved a training accuracy of 99.95% and testing accuracy of 99.66%, while InceptionV3 reached 99.85 for training and 99.39% for testing. However, DENS-INCEP outperformed both, achieving 99.96% training and 99.86% testing accuracy. This is thanks to its enhanced

ability to extract a broader range of relevant features, making it more effective at identifying subtle differences between rice varieties. The superior performance of DENS-INCEP can be attributed to this unique combination of architectures. DenseNet-201, pre-trained on the ImageNet dataset, offers robust and diverse features, while the Inception module allows the model to better capture features at different scales. Together, they enable DENS-INCEP to effectively differentiate between rice varieties.

As seen in Table 5 highlights the classification results, where the model performed exceptionally well. For instance, Ipsala achieved perfect classification (100% accuracy, sensitivity, and specificity). Although some misclassifications occurred with varieties like Jasmine and Arborio, the accuracy remained high overall. The confusion matrix in Figure 11 shows that only 14 samples of Jasmine were misclassified as Arborio, and 8 samples of Karacadag were misclassified. These minor errors can be attributed to visual similarities between some varieties. Additionally, the ROC curve in Figure 12 demonstrates the model's excellent performance, with an AUC of 1.00 for all five classes, confirming its ability to accurately distinguish between the rice varieties. This additionally reinforces DENS-INCEP's high sensitivity and specificity in classification tasks.

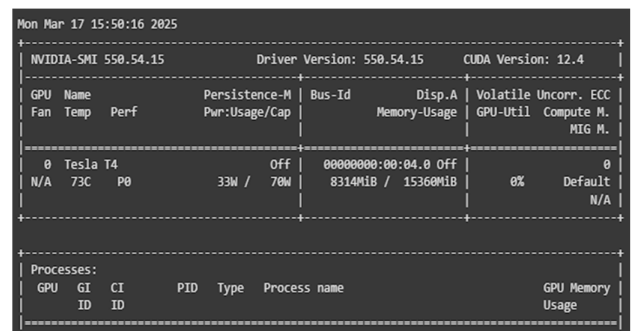


FIGURE 14. Results of computational efficiency.

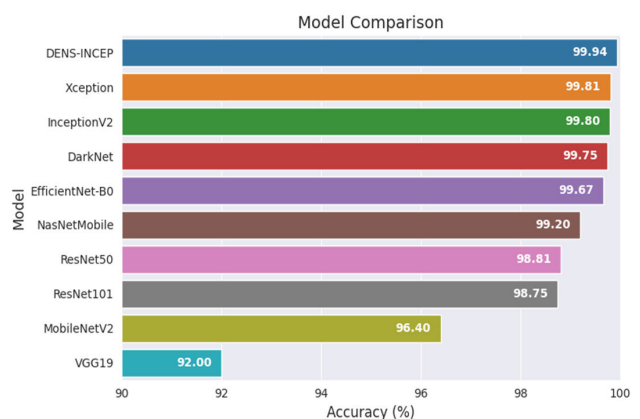


FIGURE 15. Benchmarking results of deep learning models.

The performance evaluation of the DENS-INCEP model reveals its efficiency in resource utilization, making it suitable for practical deployment. Results show that the model utilizes approximately 8.3 GB of GPU memory out of the 15.36 GB available on the Tesla T4 GPU, accounting for 54% of total GPU capacity as shown in Figure 14. This demonstrates that the model effectively manages memory, allowing deployment on systems with limited GPU resources. Additionally, the results indicate an average inference time of 0.22 seconds per batch, with 352 batches processed in 78 seconds. This low latency highlights the model's suitability for real-time inference applications requiring rapid classification. Moreover, findings confirm that the DENS-INCEP model's compact size of 89.5 MB facilitates seamless deployment on devices with limited storage, eliminating the need for extensive hardware resources.

The benchmarking results, as illustrated in Figure 15, highlights the superior performance of DENS-INCEP proposed in our study, achieving the highest classification accuracy of 99.94%. This confirms its effectiveness in extracting multi-scale shape-related features, making it the most suitable model for rice variety identification in this study. Among the comparative studies using Xception [30] and InceptionV2 [29] with accuracy 99.81% and 99.80% closely follow DENS-INCEP, demonstrating strong classification capabilities due to their efficient depthwise separable convolutions and inception modules. Other studies employing DarkNet [31] and EfficientNet-B0 [6] also achieved high accuracy (99.75% and 99.67% respectively), reinforcing the importance of optimized architectures in enhancing feature learning.

Other models, including NasNetMobile (99.20%) [28], ResNet50 (98.81%) and ResNet101 (98.75%) [33], performed well but fell slightly behind, likely due to differences in their ability to capture fine-grained shape details. Meanwhile, MobileNetV2 (96.40%) and VGG19 (92.00%) investigated by Vania et. al [2] and Cheng et. al [32] showed comparatively lower accuracy, indicating that lightweight models and older architectures may not be as effective in classifying rice varieties with subtle shape variations. These results underscore that while several deep learning models demonstrate competitive accuracy, DENS-INCEP outperforms them by effectively leveraging both DenseNet-201 for deep feature extraction and the Inception module for multi-scale shape analysis. Consequently, this study confirms DENS-INCEP's potential as the optimal model for rice variety classification, particularly in datasets where shape-related features play a critical role.

V. CONCLUSION

Efficiency in rice variety identification is crucial for improving production, quality control, and food distribution. Traditional manual methods, relying on visual inspection, often suffer from subjectivity and human error, making them less viable for large-scale processing. Therefore, there is a pressing need for an automated, accurate identification method

to address these challenges. TL offers a promising solution by leveraging pre-trained models that have already learned complex features from large datasets, thus reducing the need for extensive training on specific tasks.

The DENS-INCEP model, which combines DenseNet201 with the Inception module, enhances the model's ability to perform multi-scale analysis and deep feature extraction. This unique combination improves accuracy and efficiency by capturing a broader range of features, enabling the model to identify subtle differences among rice seed varieties more effectively. Additionally, the integration of DenseNet-201's dense connections with the Inception module allows the model to process information more efficiently, reducing computational complexity without sacrificing performance.

The experimental results from this study underscore the effectiveness of the DENS-INCEP model. It achieved outstanding performance, with an average accuracy of 99.94% across individual rice varieties, demonstrating its ability to correctly classify various rice seeds. Moreover, the model delivered excellent Sensitivity and Specificity values of 99.86% and 99.96%, respectively, further reinforcing its capability to reliably identify and differentiate rice varieties. These results suggest that DENS-INCEP is a highly promising method for practical applications in agriculture, providing a reliable and efficient alternative to traditional rice seed identification methods.

While the DENS-INCEP model has demonstrated high accuracy in shape-based rice variety classification, certain limitations must be acknowledged to guide future research. One key limitation is that the model was evaluated using a controlled dataset, which may not fully capture the variability found in real-world agricultural settings, where factors such as lighting conditions, image capture angles, grain impurities, and environmental variations could impact classification performance. Future research should focus on testing and adapting the model to diverse, field-based datasets including real-world environmental variations to ensure robust generalization in classifying shape-related features across different rice production environments. Furthermore, optimizing the model for lightweight deployment on edge devices and mobile platforms could increase accessibility for on-site seed quality assessment and automated classification in agricultural industries.

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